Reviewer Pack — Noncommutative L^p Norm of Communication Matrix Bounds Rando…

Entry cfd6712a7d9a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-12 06:43:15 UTC

Noncommutative L^p Norm of Communication Matrix Bounds Randomized Disjointness Complexity

Entry ID: `cfd6712a7d9a` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-12 06:43:15 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative L^p Geometry **Field B** (complexity object): Communication Complexity of DISJOINTNESS

Statement:

> For a random instance M of DISJOINTNESS on n elements, the noncommutative L^p norm ||M||_{p} ≥ Ω(√n) for p=2, with equality achieved when M is a balanced bipartite graph with uniform edge weights.

Rationale (proposer's reasoning):

> Noncommutative L^p norms capture operator-theoretic structure of tensors, which may encode entanglement-like constraints in communication protocols. The L^2 norm's relation to matrix rank could expose hidden geometric barriers in randomized protocols for DISJOINTNESS.

Taxonomy category: `GCT` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `8b1f5b37a2e89ba8`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM_1	LLM_2
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    n = 40
    p = 2
    target_metric_value = math.sqrt(n)
    threshold = 0.95 * target_metric_value

    random.seed(seed)

    def generate_balanced_bipartite_graph(n):
        A = [[0] * n for _ in range(n)]
        for i in range(n // 2):
            for j in range(n // 2, n):
                A[i][j] = 1
                A[j][i] = 1
        return A

    def frobenius_norm(matrix):
        norm = 0.0
        for row in matrix:
            for val in row:
                norm += val ** 2
        return math.sqrt(norm)

    metric_value = 0.0
    instances_tested = 30

    for _ in range(instances_tested):
        M = generate_balanced_bipartite_graph(n)
        metric_value += frobenius_norm(M)

    mean_metric_value = metric_value / instances_tested
    conjecture_holds = mean_metric_value >= threshold
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "Frobenius Norm",
```

```
        "metric_value": mean_metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if len(sys.argv) > 1 else [601, 631, 677, 727, 773, 821

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if not results:
        raise ValueError("No trials were executed. Ensure seeds are provided.")

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=undefined support_fraction={support
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_f
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
c_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30, 'conjecture_h
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
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TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 28.284271247461902, 'instances_tested': 30,
RESULT: SUPPORTED mean=28.284271247461902 std=undefined support_fraction=1.0
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

Safety rail: critic_challenge | original judge: | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	74539
2	preregistration	ollama_remote	qwen3:8b	0	0	24740
3	novelty	ollama_remote	qwen3:8b	0	0	20835
4	novelty	ollama_remote	qwen3:8b	0	0	11572
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12745
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7016
7	critic	ollama_remote	qwen3:8b	0	0	40922
8	judge	ollama_remote	qwen3:8b	0	0	21402

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 213772 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/cfd6712a7d9a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/cfd6712a7d9a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/cfd6712a7d9a.tar.gz` (if generated)